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Molecular and Genomic Approaches to Clinical Microbiology in Africa

A glimpses into the World of MRC Unit The Gambia

Professor Martin Antonio PhD, FRCP, FRCPath, FAAS

MRC Investigator, Medical Research Council Unit The Gambia @

London School of Hygiene & Tropical Medicine

Co-Director LSHTM Centre for Epidemic Preparedness and Response

Professor of Molecular Microbiology and Global Health.

Sunday 13th Sep 2025



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Target Malaria activities suspended in Burkina Faso

Posted on August 22, 2025



By Target Malaria

Target Malaria is an innovative project aiming to reduce the population of malaria-transmitting mosquitoes in sub-Saharan Africa.

Burkina Faso ends flagship malaria project

BURKINA FASO | BURKINA FASO



APA-Ouagadougou (Burkina Faso) — 22 August 2025 | 18:42



A Vector Control Research Alliance

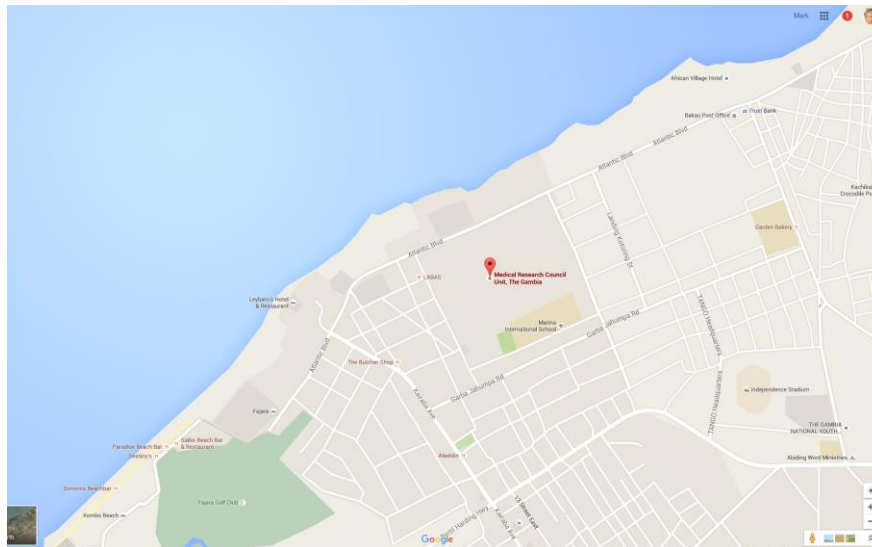
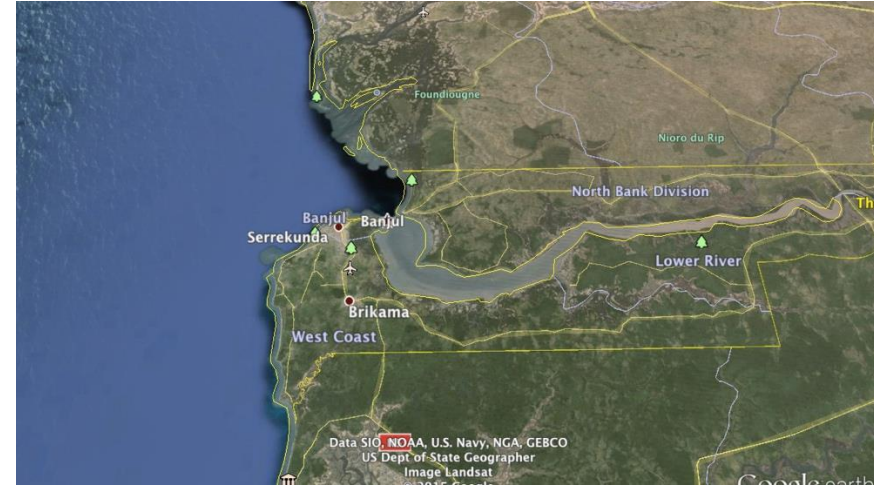


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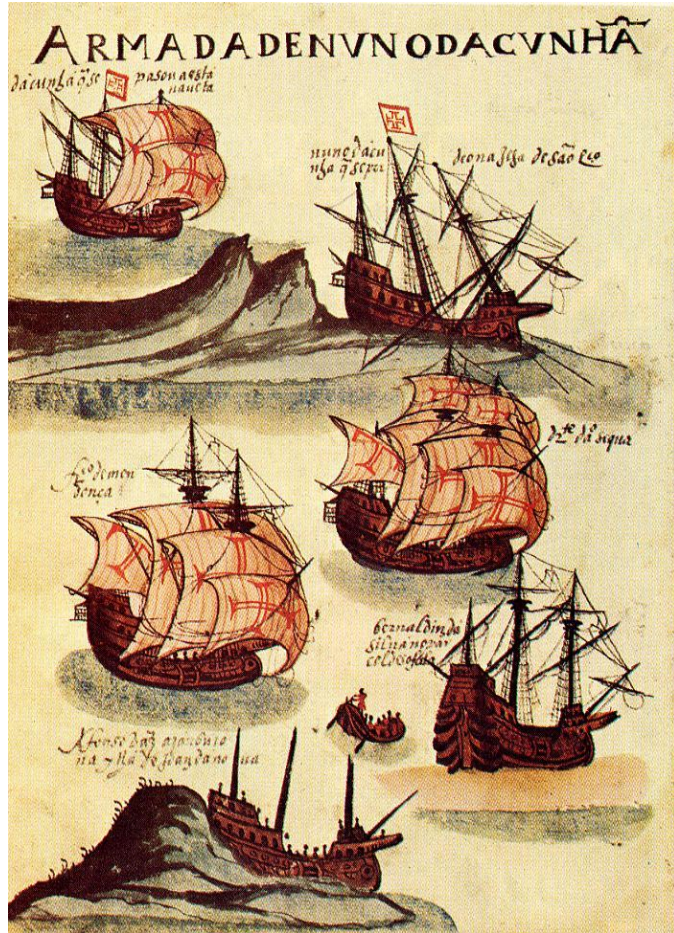


The Gambia



ARRIVAL OF THE EUROPEANS:

Gambia first under British control 1765



Gambia River reached in 1446 by Dinis Dias



Portuguese map of 1502

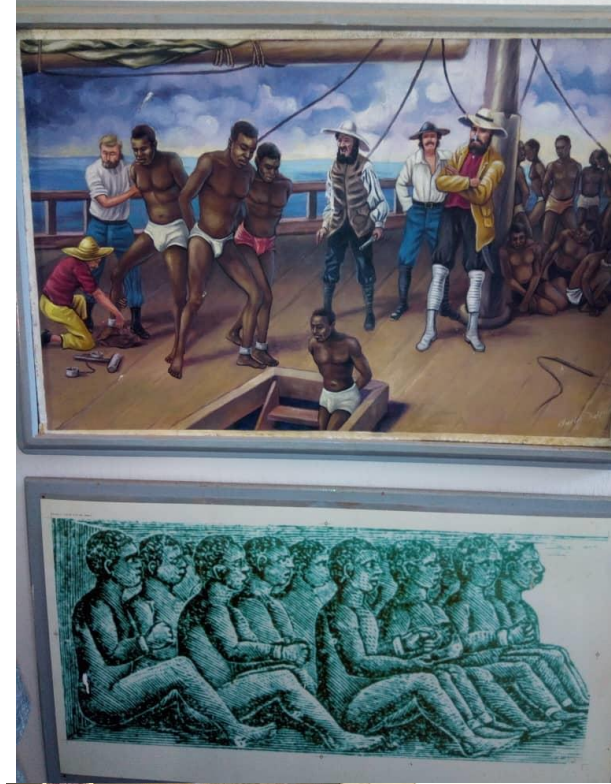
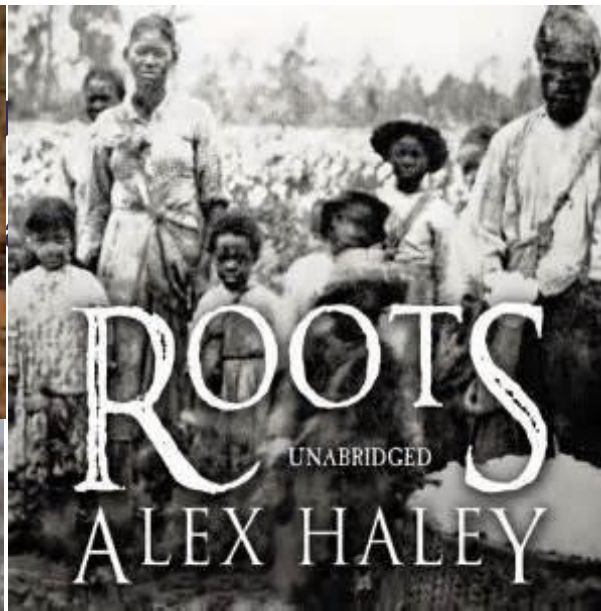
1. Portuguese discovered River Gambia 1446 – trade to the interior; raids
2. 1587, British merchants traded
3. 1678 Royal African Company fort on James Island
4. Gambia first under British control 1765

Gambian history in leaps

Kunta Kinteh {St James}
Island

https://en.wikipedia.org/wiki/Kunta_Kinteh_Island#History

CONDITIONS OF
SLAVERY





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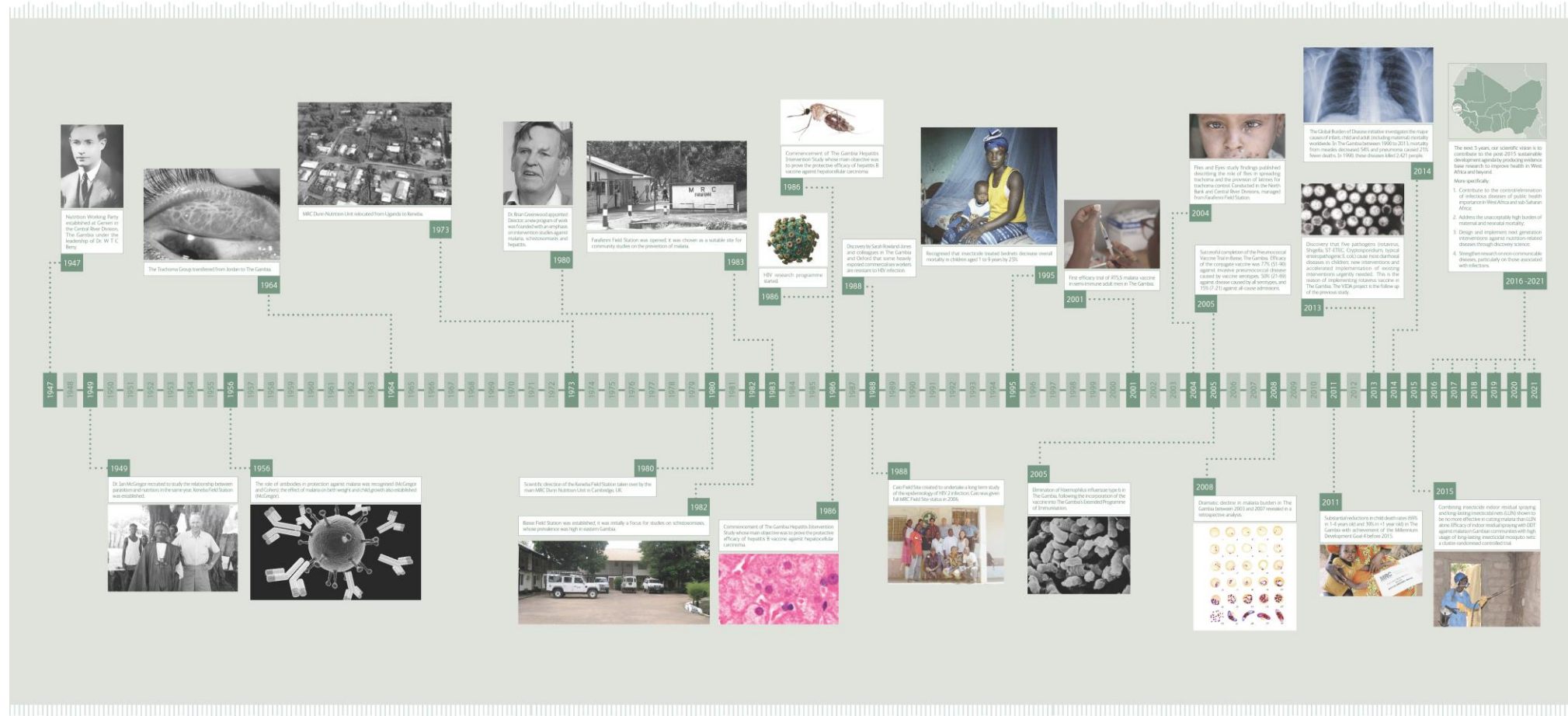
Independence Day: 1965





MRC Unit The Gambia: est. 1947

MRC | Unit
The Gambia





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History of the MRC Unit The Gambia: est. 1947





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MRC Gambia - Some History

- 1945 – MRC first reports from Gambia, nutrition research party
- 1948 - Dr Ian McGregor studies parasites and nutrition; Keneba Field Station was established
- 1954 - Dr I McGregor appointed Director, wider scope of unit
- 1964 - The role of antibodies in protection against malaria established (McGregor & Cohen); the effect of malaria on birth weight and child growth established (McGregor)



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Early Tropical Medicine (1901) MRC The Gambia Unit: Est 1947



**Contributed to elimination
of major health problems
in West Africa (including
The Gambia)?**



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Training team



[Kanny Diallo](#)
Centre Suisse de
Recherche Scientifique



[Brenda Kwambana-Adams](#)
Liverpool School of
Tropical Medicine and
the Malawi-Liverpool-
Wellcome Clinical
Research Programme



[Martin Maiden](#)
University of Oxford



[Elizabeth Streicher](#)
University of
Stellenbosch



[Dam Khan](#)
MRC Unit The Gambia at



[Keith Jolley](#)
University of Oxford



[Leonard Ndwiga](#)
University of Glasgow



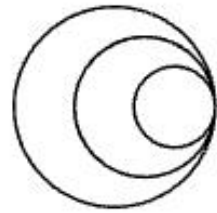
[Anzaan Diepenaar](#)
University of Antwerp

Before and After



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science**

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Molecular and Genomic Approaches to Clinical Microbiology in Africa

13–19 September 2025

MRC Unit, The Gambia

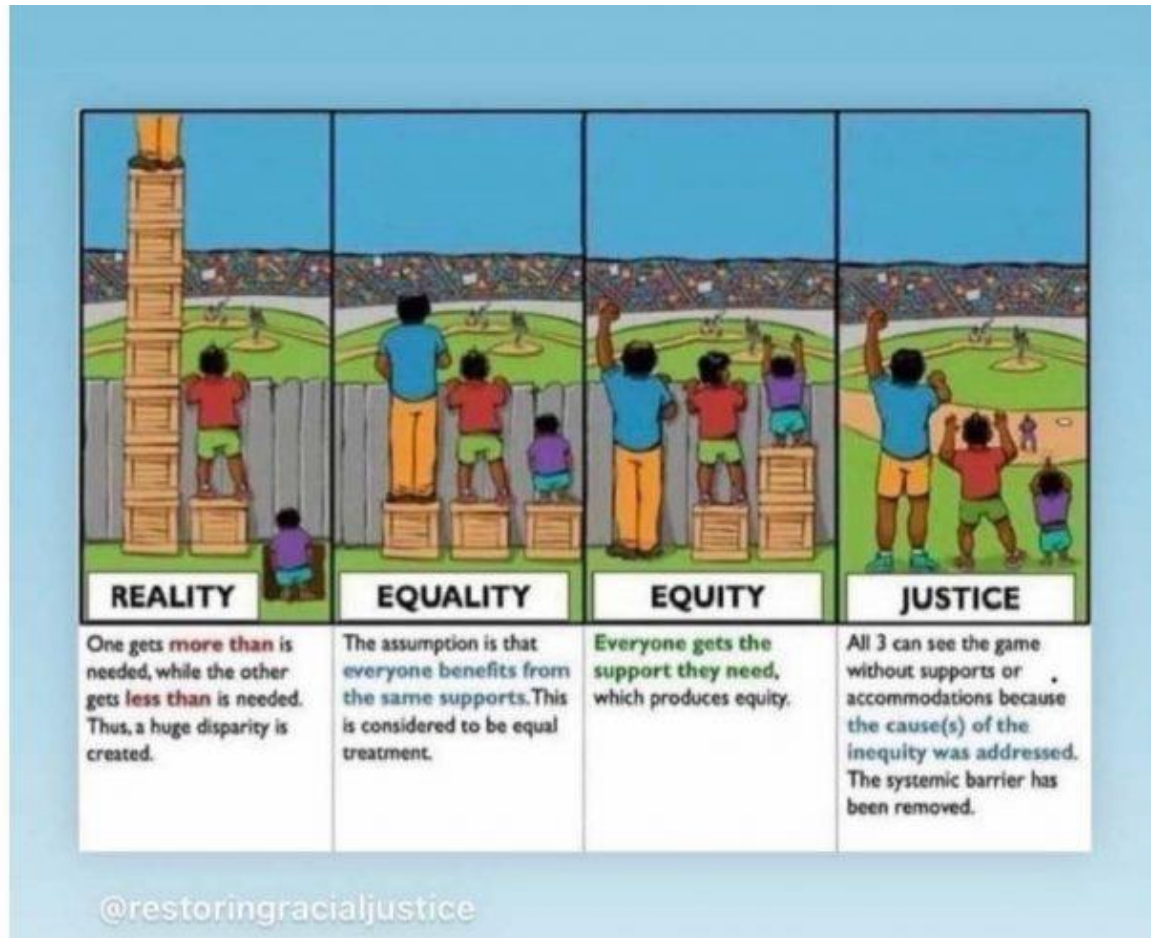


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Inequality, equality equity & justice





Current MRC Campus in Fajara





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**Leading Health Research in Africa to
Improve Lives Across the World**
Most vulnerable population, particularly
women & children





Anopheles gambiae



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MRC The Gambia Some History



1980 –

Dr Brian Greenwood
appointed Director;
intervention studies against
malaria, schistosomiasis
and hepatitis.



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Rebellion

- 1981
- Violent coup attempt
- Up to 800 dead
- Dramatic resolution – SAS and MRC





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Life goes on – Basse Field Station established 1982



1982 est- Basse field site



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Weather, geography





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MRC The Gambia more History

1986 – Commencement Gambia Hepatitis Intervention Study

1995 - Established that insecticide treated bednets decrease mortality in children 25%

1995 – completion Hib vaccine efficacy trial

2004 – Tumani Corrah Director

2005 - completion of the Pneumococcal Vaccine Trial in Basse

2005 - Near elimination of Hib in The Gambia





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PNEUMOCOCCAL VACCINATION IN THE GAMBIA



Epidemiology



Polysaccharide vaccine - children



Polysaccharide vaccine – maternal



Immunogenicity 5-valent



Booster study



Immunogenicity 7-valent



Mixing study 9 valent



Phase 3 efficacy trial 9-valent



**Implementation of
National
immunisation**



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The Pneumococcal Vaccine Trial (PVT) Story

- Pneumococcus - leading bacterial cause of pneumonia
- Gambian PVT - PCV9 (Wyeth)
- Plan – 40,000 children URR/CRR (Basse base)
- Problem 1 – Wyeth abandoned PCV9 → <20k children
- Problem 2 – near shutdown



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PVT – a Result!

Outcome	VE (95% CI)
IPD vaccine serotypes	77 (51, 90)
IPD all serotypes	50 (21, 69)
Radiological pneumonia	37 (27, 45)
Mortality all cause	16 (3,28)

Cutts *et. al.* Lancet. 2005; **365**: 1139–1146



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PCV into Gambian EPI August 2009





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The Gambia a's EPI: a product of partnership

**Hepatitis B Vaccine 1986:
1st country in Africa**

**Hib vaccine: 1st country in Africa
1997: effective long term**

**Prevnar 2009:
2nd country in Africa**

PCV-13: 2011

Handwritten: Hib/MSI Campaign 12-12-11

PERSONAL HISTORY: 2011/6105 CHILD'S NUMBER ASB-305

Child's Name:	Date of Birth: 11-3-11
Mother's Name:	Tribe or Ethnic Group:
Father's Name:	Place of Delivery:
Address/Name of Village and Compound:	Health Facility: ☒ Home: ☐
	Health Worker: ☒ TBA: ☐ Other: ☐
Welfare Clinic: HSB	Sex: Male Birth Weight: 2.6 kg
	Date First Seen: 12-3-11

BROTHERS AND SISTERS:				HOME VISITS:	
Year of Birth	Sex	Alive and Well	Age at Death	Date	Outcome

IMMUNISATION RECORD:		DATE REQUIRED	DATE RECEIVED
ANTI-TUBERCULOSIS AND HEPATITIS IMMUNISATION			
BCG Injection (at birth or soon after)		12-3-11	12-3-11
Date of recognizing BCG Scar			
Hepatitis B (at birth or soon after)		12-3-11	12-3-11
POLIOMYELITIS IMMUNISATION			
Polio 0 (at birth or soon after)		12-3-11	12-3-11
Polio 1 (at the age of 2 months)			11-5-11
Polio 2 (one month after second dose)			15-6-11
Polio 3 (one month after third dose)			18-8-11
Polio 4 (at the age of nine months or later)			
Booster (at the age of eighteen months or later)			
Pentavalent Immunisation			
Pentavalent 1 (at the age of two months or later)			11-5-11
Pentavalent 2 (one month after first injection)			15-6-11
Pentavalent 3 (one month after second injection)			18-8-11
DPT Booster (one year after third injection)			
PNEUMOCOCCAL CONJUGATE VACCINE IMMUNISATION			
Pneumo 1 (at the age of two months)			11-5-11
Pneumo 2 (one month after first injection)			15-6-11
Pneumo 3 (one month after second injection)			18-8-11
MEASLES IMMUNISATION			
Measles injection (at the age of 9 months or later)			
YELLOW FEVER IMMUNISATION			
Yellow Fever Injection (at the age of 9 months or later)			
VITAMIN A		Date Required	Date Received
1 st Dose 100,000 IU at 6 months			12-10-11
2 nd Dose 200,000 IU at 12 months			
3 rd Dose 200,000 IU at 18 months			
4 th Dose 200,000 IU at 24 months			
5 th Dose 200,000 IU at 30 months			
6 th Dose 200,000 IU at 36 months			
7 th Dose 200,000 IU at 42 months			
8 th Dose 200,000 IU at 48 months			
9 th Dose 200,000 IU at 54 months			
10 th Dose 200,000 IU at 60 months			
Post Partum Mothers Vitamin A 200,000 IU (within 2 weeks after delivery)			

Received LLIN	
Date	
Yes ☐	No ☐

(TT (MOTHERS ONLY))	
Dose	Date
TT-1	
TT-2	
TT-3	
TT-4	
TT-5	

THE GLOBAL FUND
to Fight AIDS, Tuberculosis and Malaria



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Regional Approach to Multidrug-resistant tuberculosis



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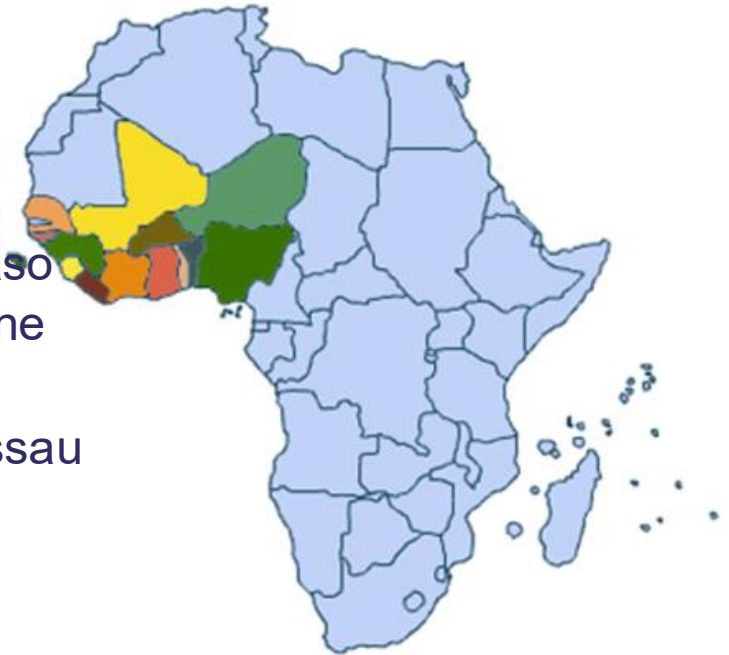


MRCG@LSHTM presence across West Africa

1. Collaborations across all research themes in the Unit.
2. Collaborations also centred around Maternal and neonate studies, and capacity building grants
3. Attraction of funds for Unit collaborations and partnerships within West Africa region is at £3.3M.
4. Collaborations within the region is centred mainly around Senegal, Ghana and Mali
5. Total number of active projects across themes being carried out within the region is 16.

Active projects: within West Africa

1. Senegal
2. Ghana
3. Nigeria
4. Mali
5. Burkina Faso
6. Sierra Leone
7. Cameroon
8. Guinea Bissau
9. Benin
10. Togo





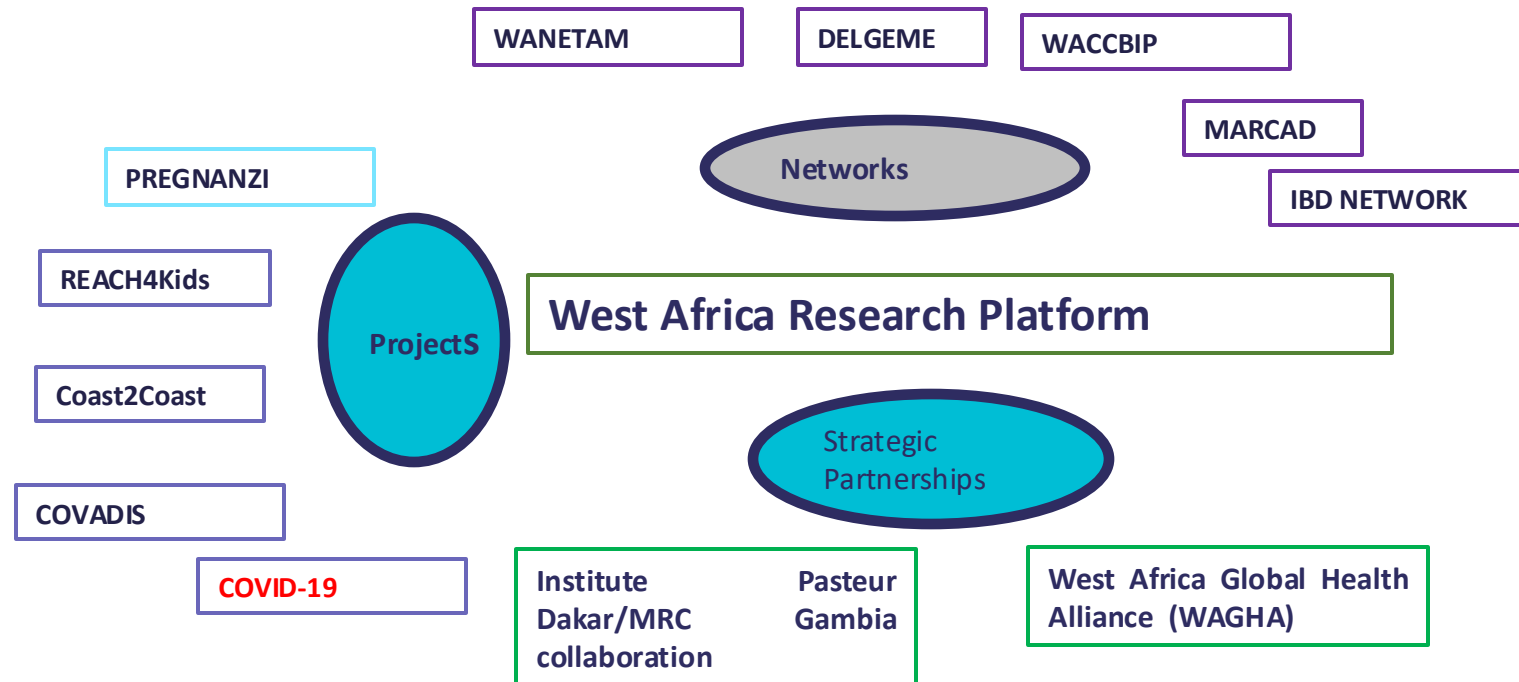
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MRC@LSHTM West Africa Research Platform

Implementation Framework



All Research Themes, Training, Genomics and Sequencing, Data Management, Biomedical Engineering, Clinical Trials
Department, Research Support Office, Biobanking



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High tech platforms close to the field (MBL)





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Tuberculosis (TB)

Yes! We Can End TB

The World Health Organization (WHO) and STOP TB Partnership **theme** for World TB Day in

2022 “Invest to End TB. Save Lives ”

2023 “Yes! We Can End TB”.

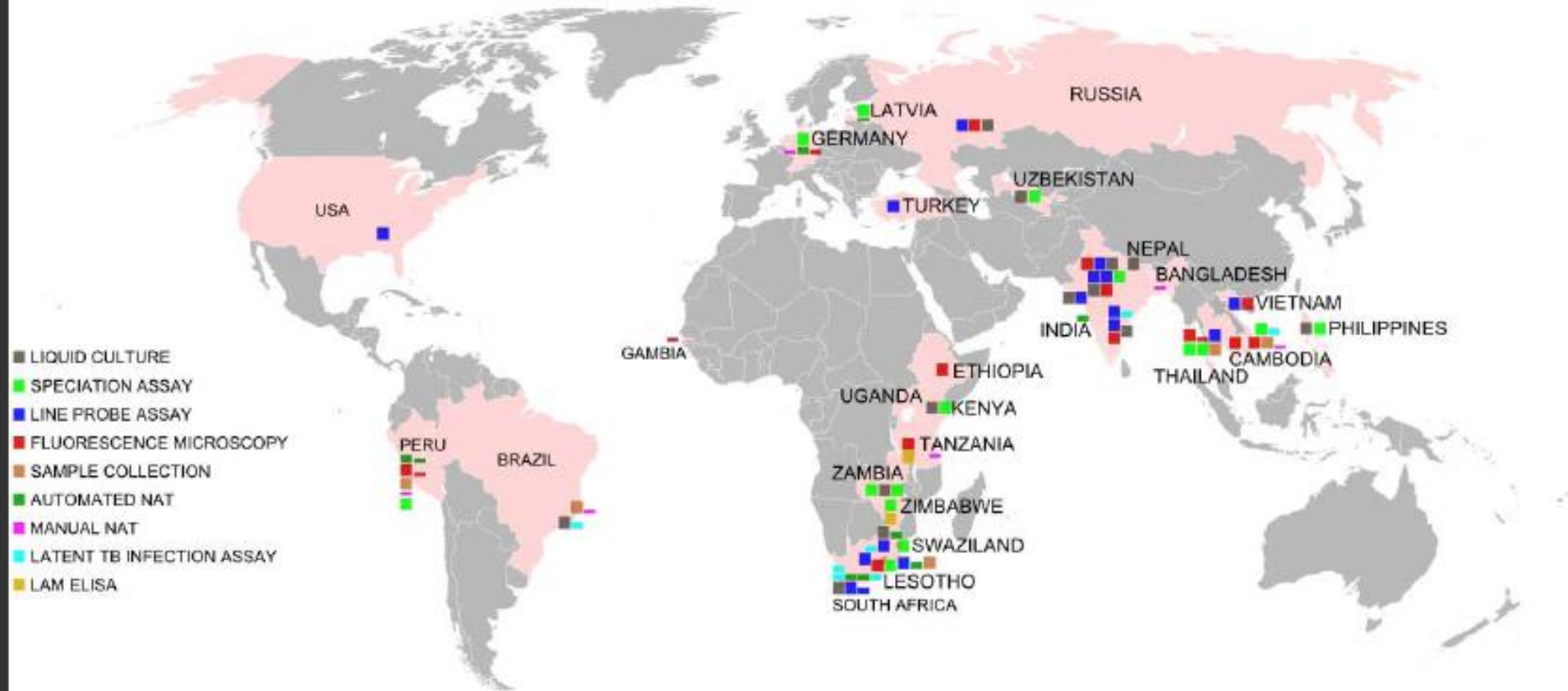
2024, the same theme “Yes! We Can End TB”

2025, Yes! We Can End TB: Commit, Invest, Deliver”

Stronger commitment is crucial, especially as declining international funding for TB in the current geopolitical climate threatens the achievement of *global targets to end TB by 2030*

Partner sites for TB trials

Where are the gaps?



- ❖ Trial infrastructure in Western and Central Africa
- ❖ Trial site networks for high-quality early phase trials
- ❖ Local repositories with well-characterized specimens
- ❖ Specific trial site networks for pediatric & extrapulmonary TB



EDCTP TB clinical trials

- Ethiopia
- Guinea-Bissau
- Kenya
- Madagascar
- Mozambique
- South Africa
- Tanzania
- Uganda
- Zambia
- Zimbabwe

- Clinical Trials - all PRD's
- Clinical Trials - HIV
- Clinical Trials - Malaria
- Clinical Trials - TB
- Training Awards
- Ethics Capacity Strengthening
- Regulatory Capacity Strengthening
- Networks of Excellence





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WANETAM-I Strategy

Strong Institution

Mentorship, training, IT
transfer of techniques,
Data management

Sister
Institution

Sister
Institution

Lab infrastructure

1. Renovation of laboratories
2. Improve research environment
e.g. biosafety, EQA
3. Epidemiological studies etc

Human resources

Short and long term (e.g.
MSc, PhD) training of
lab technicians, researchers, physicians,
bio-medical engineering,
administrative etc



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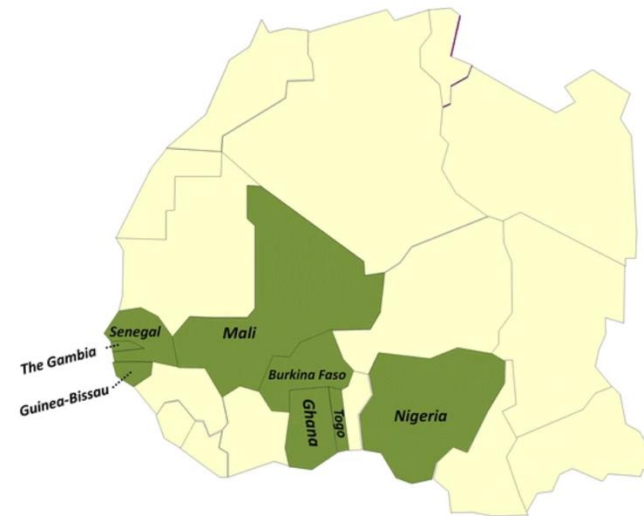
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Participating WANETAM 1 TB sites

Coordinated by *MRC*, The Gambia, the following partner sites were part of the capacity building activities and drug resistance surveys:

- (1) Senegal, Laboratoire Bactériologie Virologie, Le Dantec, Dakar;
- (2) The Gambia, National Public Health Laboratory Services, Banjul;**
- (3) Guinea-Bissau, National Institute of Public Health (*INASA*), Bissau;
- (4) (Mali, SEREFO (HIV/TB Research and Training Center) FMOS, University of STT, Bamako;
- (5) Burkina Faso, Centre Muraz and the National TB Program (*NTP*), Ouagadougou;
- (6) Ghana, Korle Bu Teaching Hospital, Accra;
- (7) Togo, Laboratoire National de Référence (*LNR*) des Mycobactéries, Lome;
- (8) Nigeria, Nigerian Institute of Medical Research (*NIMR*), Lagos;
- (9) Nigeria, College of Medicine, University of Ibadan



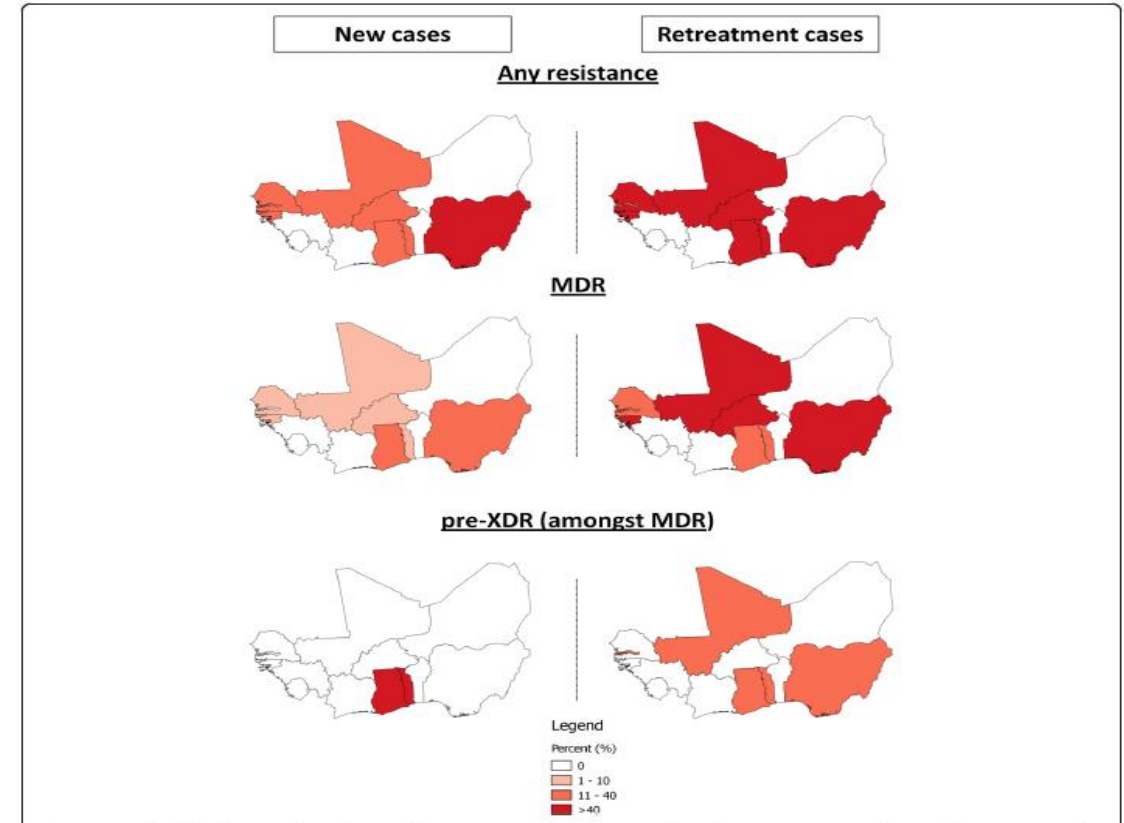
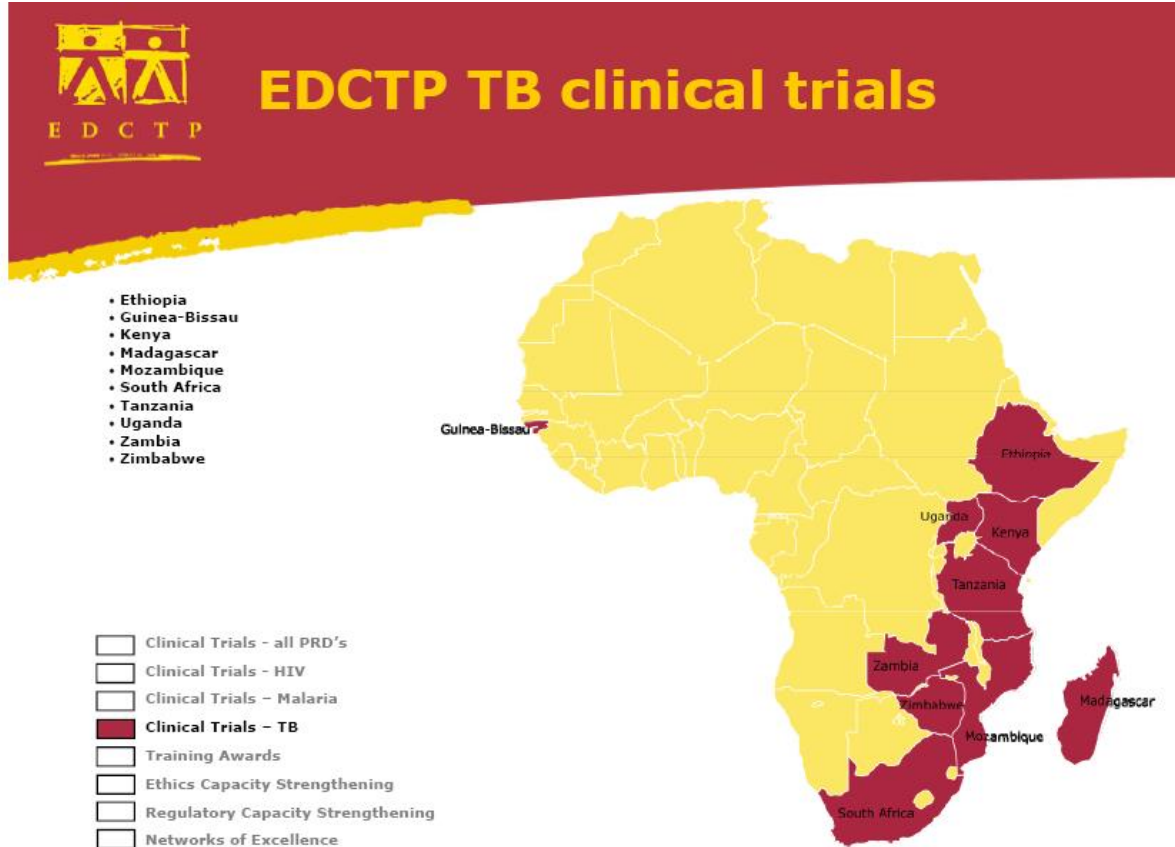


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TB Clinical Trials sites in West Africa



Before and after



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Comparison between WANETAM (blue) and WHO Global TB report, 2014 (red) MDR prevalence estimates.

Gehre *et al. BMC Medicine* (2016) 14:160
DOI 10.1186/s12916-016-0704-5

BMC Medicine

RESEARCH ARTICLE

Open Access

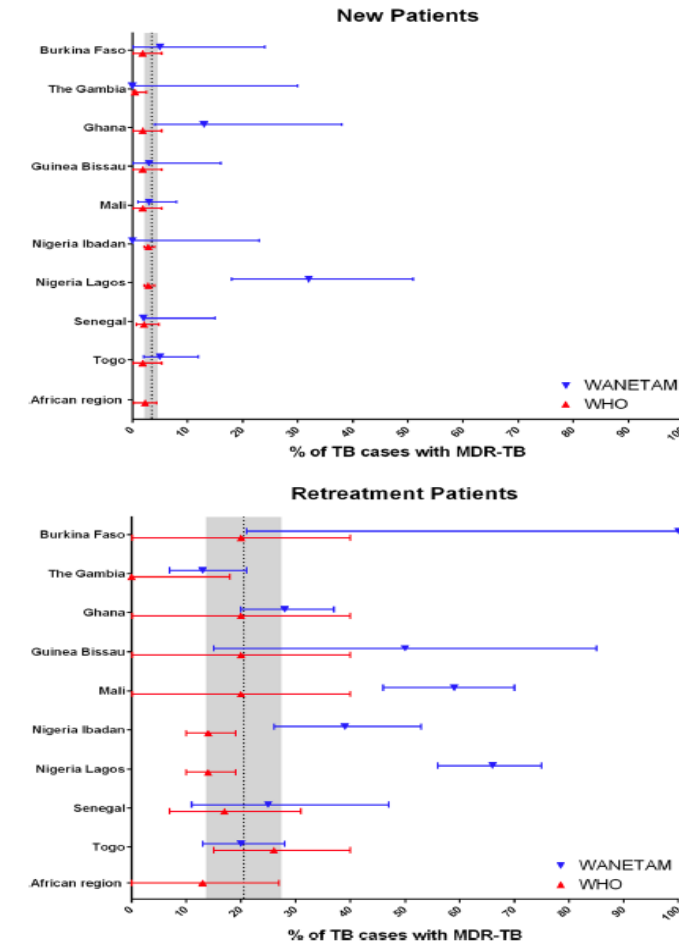


The emerging threat of pre-extensively drug-resistant tuberculosis in West Africa: preparing for large-scale tuberculosis research and drug resistance surveillance

Florian Gehre^{1,19†}, Jacob Otu^{1†}, Lindsay Kendall^{1†}, Audrey Forson^{2,3}, Awewura Kwara^{4,5}, Samuel Kudzawu³, Aderemi O. Kehinde^{6,7}, Oludele Adebiyi⁷, Kayode Salako⁷, Ignatius Baldeh⁸, Aisha Jallow⁸, Mamadou Jallow⁸, Anoumou Dagnra⁹, Kodjo Dissé⁹, Essosimna A. Kadanga⁹, Emmanuel Oni Idigbe¹⁰, Catherine Onubogu¹⁰, Nneka Onyejebu¹⁰, Aissatou Gaye-Diallo¹¹, Awa Ba-Diallo¹¹, Paulo Rabna¹², Morto Mane¹², Moumine Sanogo¹³, Bassirou Diarra¹³, Zingue Dezemon¹⁴, Adama Sanou¹⁴, Madikay Senghore^{1,15}, Brenda A. Kwambana-Adams^{1,16}, Edward Demba¹, Tutty Faal-Jawara¹, Samrat Kumar¹⁹, Leopold D. Tientcheu^{1,22}, Adama Jallow²³, Samba Ceesay²⁴, Ifedayo Adetifa^{17,18}, Assan Jaye¹, Mark J. Pallen¹⁵, Umberto D'Alessandro^{16,18,19}, Beate Kampmann^{1,20}, Richard A. Adegbola²¹, Souleymane Mboup¹¹, Tumani Corrah¹, Bouke C. de Jong¹⁹ and Martin Antonio^{1,15,17*}

WHO estimated the prevalence of MDR-TB in new and retreatment cases in the sub-region to be 2% and 17%

WANETAM found 6% and 35% respectively (Gehre *et al.*, 2016)





RESEARCH ARTICLE

Open Access



The emerging threat of pre-extensively drug-resistant tuberculosis in West Africa: preparing for large-scale tuberculosis research and drug resistance surveillance

Florian Gehre^{1,19†}, Jacob Otu^{1†}, Lindsay Kendall^{1†}, Audrey Forson^{2,3}, Awewura Kwara^{4,5}, Samuel Kudzawu³, Aderemi O. Kehinde^{6,7}, Oludele Adebisi⁷, Kayode Salako⁷, Ignatius Baldeh⁸, Aisha Jallow⁸, Mamadou Jallow⁸,



RESEARCH ARTICLE

Mycobacterium africanum (Lineage 6) shows slower sputum smear conversion on tuberculosis treatment than *Mycobacterium tuberculosis* (Lineage 4) in Bamako, Mali

Bassirou Diarra^{1,2,3*}, Mahamadou Kone¹, Antieme Combo Georges Togo¹, Yeya di Sadio Sarro¹, Aissata Boubakar Cisse⁴, Amadou Somboro¹, Boureima Degoga¹, Mohamed Tolofoudie¹, Bourahima Kone¹, Moumine Sanogo¹, Bocar Baya¹, Ousmane Kodio¹, Mamoudou Maiga⁵, Michael Belson⁵, Susan Orsega⁵, Mervam Krit

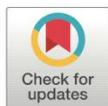


RESEARCH ARTICLE

Whole-genome sequencing illuminates the evolution and spread of multidrug-resistant tuberculosis in Southwest Nigeria

Madikay Senghore^{1,2}, Jacob Otu¹, Adam Witney³, Florian Gehre^{1,4}, Emma L. Doughty², Gemma L. Kay², Phillip Butcher³, Kayode Salako⁵, Aderemi Kehinde⁵, Nneka Onyejebu⁶, Emmanuel Idigbe⁶, Tuman Corrah¹, Bouke de Jong⁴, Mark J. Pallen^{2,7}, Martin Antonio^{1,2,8*}

1 Vaccines and Immunity Theme, Medical Research Council Unit The Gambia, Fajara, The Gambia, **2** Microbiology and Infection Unit, The University of Warwick, Coventry, United Kingdom, **3** Institute of Infection and Immunity, St George's University of London, London, United Kingdom, **4** Institute of Tropical Medicine, Antwerp, Belgium, **5** Department of Medical Microbiology & Parasitology, University College Hospital, Ibadan, Nigeria, **6** National Tuberculosis Reference Laboratory, Nigeria Institute of Medical Research, Lagos, Nigeria, **7** Quadram Institute, Norwich Research Park, Norwich, Norfolk, NR4 7UA, **8** Faculty of Infectious and Tropical Diseases, London School of Hygiene & Tropical Medicine, London,



SCIENTIFIC REPORTS

OPEN

Comparative genomics of *Mycobacterium africanum* Lineage 5 and Lineage 6 from Ghana suggests distinct ecological niches

Isaac Darko Otchere^{1,2}, Mireia Coscollá^{3,4}, Leonor Sánchez-Busó⁵, Adwoa Asante-Poku¹, Daniela Brites^{3,4}, Chloe Loiseau^{3,4}, Conor Meehan⁵, Stephen Osei-Wusu¹, Audrey Forson⁷,

Received: 23 November 2017
Accepted: 16 July 2018
Published online: 26 July 2018



ORIGINAL RESEARCH

Second-line anti-tuberculosis drug resistance testing in Ghana identifies the first extensively drug-resistant tuberculosis case

Stephen Osei-Wusu^{1,2}, Michael Amo Omari³, Adwoa Asante-Poku¹, Isaac Darko Otchere¹, Prince Asare¹, Audrey Forson³, Jacob Otu⁴, Martin Antonio⁴

Background: Drug resistance surveillance is crucial for tuberculosis (TB) control. Therefore, our goal was to determine the prevalence of second-line anti-TB drug resistance among diverse primary drug-resistant *Mycobacterium tuberculosis* complex (MTBC) isolates in Ghana.

Materials and methods: One hundred and seventeen MTBC isolates with varying first-line drug resistance were analyzed. Additional resistance to second-line anti-TB drugs (streptomycin [STR], amikacin [AMK] and moxifloxacin [MOX]) was profiled using the Etest and GenoType MTBDRsl version 2.0. Genes associated with resistance to AMK and MOX (*gyrA*, *gyrB*, *eis*, *rrs*, *tap*, *whiB7* and *tlyA*) were then analyzed for mutation.

Results: Thirty-seven (31.9%) isolates had minimum inhibitory concentration (MIC) values

Contribution of Xpert[®] MTB/RIF to the diagnosis of pulmonary tuberculosis among TB-exposed children in The Gambia

Authors: Togun, T. O.¹; Egere, U.¹; Sillah, A. K.¹; Ayorinde, A.¹; Mendy, F.¹; Tientcheu, L.¹; Otu, J.¹; Antonio, M.¹; Sutherland, J.¹; Hill, P. C.²; Kampmann, B.³

Source: The International Journal of Tuberculosis and Lung Disease, Volume 19, Number 9, 1 September 2015, pp. 1091-1097(7)

Publisher: International Union Against Tuberculosis and Lung Disease

DOI: <https://doi.org/10.5588/ijtld.15.0228>

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TB WANETAM II

- **Childhood TB:** Improved diagnosis and management of TB in children
 - how to find the children who are ill with tuberculosis (TB) but never get diagnosed?
- **Drug resistant TB:** Efficient diagnosis and management of drug-resistant TB
 - new tools for improving the diagnosis, treatment and prevention of TB



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2011-2015

Enrolled **4,232 cases** and **5,325 controls**

- **Specimen type:** Induced sputum or Gastric aspirate; lung aspirates; or pleural fluid
- MRC The Gambia **recruited 700 children**
- **1 case/700 (0.14%)** of microbiological confirmed TB (Sputum smear + culture)

Novel diagnostics for TB (especially childhood TB). Xpert MTB/RIF Ultra



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WANETAM-3 DR-TB Group



Prof. Martin Antonio (MRCG@LSHTM, The Gambia)

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Aim 1: To determine the burden of drug resistance TB in the West African Sub-region. (WANETAM Stes)

Objective 1: To isolate bacteria from sputum of GeneXpert-positive TB patients and genotype the isolates using PCR amplification of IS6110 and MTBC-specific *rpoB*

Objective 2: To carry out sequential phenotypic screening of obtained isolates for drug resistance against DR-TB strains

Aim 2: To determine the genetic drivers of the emergence and transmission of DR-TB in West Africa (MRC Fajara site)

Objective 1: To whole genome sequence prospective and archived DR-TB isolates using the illumine platform

Objective 2: To carry out phylogenetics and comparative genomics analysis of DR-MTBC genomes including *in silico* mutational analysis



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Defying barriers to fight tuberculosis in West Africa: a model of equitable partnerships within a research capacity-strengthening network in the subregion

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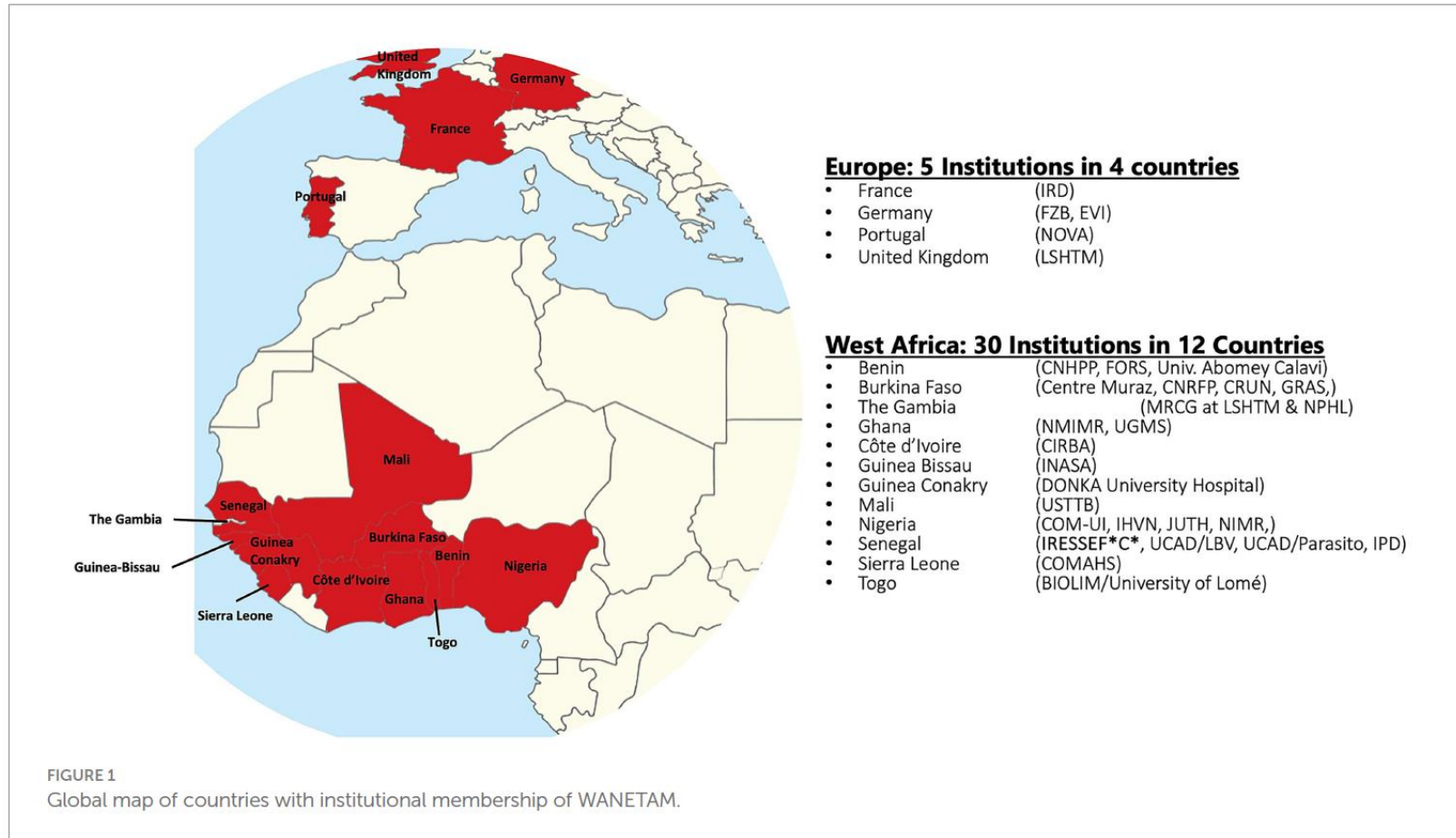


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TABLE 1 Research and administrative support by WANETAM to West African institutions.

Area of improvement	Name of site	Level at baseline	Current level
Molecular Biology	UCAD HIV	2	WHO-accredited
Biobank	UCAD HIV	1	3
Clinical Trial	UCAD HIV	1	3
Chest Clinic, TB Laboratory	UGMS, Ghana	1	2
TB Reference Laboratory	BHP, Guinea-Bissau	1	3
Laboratories	University of Ibadan, Nigeria	1	3
Reference Laboratory	NIMR, Nigeria	2	4
Clinical trial facilities	UCAD Coordination, Senegal	2	4
	UCAD HIV/TB Senegal	2	4
	University of Ibadan, Nigeria	1	3
	Innovative Biotech, Nigeria	1	3
Data management	UCAD Coordination Senegal	1	4
	UCAD HIV/TB, Senegal	1	4
	University of Ibadan, Nigeria	1	4
Information technology and data sharing facilities	UCAD Coordination, Senegal	2	4
	UCAD HIV/TB Senegal	2	4
	KBTH, Ghana	1	2
	BGP, Guinea-Bissau	1	4
	University of Ibadan, Nigeria	1	4
Finance and administration	UCAD Coordination Senegal	2	4
	UCAD HIV/TB Senegal	2	4
	KBTH, Ghana	1	2
	University of Ibadan, Nigeria	2	4

The grading system used in the table corresponds to the SLIPTA star system, where “1, 2,3, and 4” represent “1 star, 2stars, 3 stars, and 4 stars,” respectively.



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TABLE 2 Varying burden of tuberculosis among WANETAM partner countries in West Africa.

Partner country	TB Deaths (thousand)		Estimated TB incidence/100,000/Year		Current population (millions)
	2009	2023	2009	2023	
Benin	1.50	1.20	69	51	13.3
Burkina Faso	8.70	0.71	60	43	22.7
Cote d'Ivoire	18.0	6.30	207	119	28.2
Gambia	0.83	0.47	180	142	2.7
Ghana	11.0	9.90	187	129	33.5
Guinea Bissau	0.49	1.60	361	361	2.1
Guinea Conakry	7.30	2.30	190	175	13.9
Mali	11.0	1.50	64	48	22.6
Nigeria	110	64.0	219	219	218.5
Senegal	9.00	1.80	134	110	17.3
Sierra Leone	9.00	1.30	318	283	8.6
Togo	7.50	0.08	77	30	8.8
Total WANETAM	194.32	91.16	172.2	142.5	392.2
Africa	430	291	333	206	824
Global	1,300	1,080	137	134	6,826

According to the 2010 and 2024 Global tuberculosis reports (4, 29) and the World Bank report (2000–2023).

FW: Health impact / networking activity - Gambia Unit case studies

☺ ↶ ↷



Umberto D'Alessandro <Umberto.Dalessandro@lshtm.ac.uk>

Thursday, 11 September 2025 at 09:30

To: Leadership

Retention: Staff mailbox default delete after 7 years Expires: 09/09/2032.

I hope you are well. I am just following up my emails below for your help please in creating short case studies to showcase as part of the MRC/FCDO concordat annual review (you will recall FCDO contributes to the Unit's core funding).

It would be great if you could provide any good examples from the Gambia Unit (ideally from the last 12 months or so) to demonstrate any of the below:

- A high quality in-depth impact case study - showing the local impact of the project during the active phase and/or the international impact of the research findings from prior research (ie translating research into health impact).
- A networking activity case study with external partners perhaps a network that the unit participates in. We are particularly interested in quantitative information such as the number of attendees etc and any impact from the activity.

If you are able to get back to me by Monday 15th September that would be most appreciated.

Many, many thanks and please do let me know if any queries.



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Conclusion: Regional Approach to Multidrug-resistant tuberculosis

- MDR-TB remains a major global health issue, particularly in West Africa, which carries 24% of the global burden and harbors diverse MTBC lineages.
- MDR-TB threatens TB control; WA may underestimate the true prevalence.
- The WANETAM TB work package focuses on
 - Evaluating diagnostic tools to ensure they are suitable for West Africa's unique MTBC diversity,
 - Conducting DR-TB surveillance to determine the existing burden of DR-TB in the sub-region,
 - Training and capacity building research leadership and technical expertise with emphasis on gender equality.
- Research conducted by members of TB working group of WANETAM has contributed to understanding prevalence and better control of TB including DR-TB in the sub-region. \
- WANETAM brings Francophone, Anglophone and Lusophone West Africans to successfully work together in fighting TB and preparing for unexpected future pandemics in the sub-region,
- **WANETAM has shown that Yes, we can stop TB, if we work together.**



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Sustainability

**Future African Leaders
PhD and Network grants**



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WANETAM TB-4



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What can possibly go wrong?

...time and chance happen to them all Eccl. 9:11

- Civil dangers (coup)
- Programme shutdowns (Trump, USAID, ODA funding)
- Expulsion (Jammeh)
- Epidemic (Covid)
- Weather, geography (seasonality)



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Problems
Purpose
People
Partnership
Privilege

